Macroeconomía Internacional

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Modelo Standard de RBC para una Pequeña Economía Abierta

The SOE-RBC model

Brief description

- Small and open economy.
- Representative household derives utility from consumption and leisure.
- Technology uses capital and labor to produce final goods, which can be traded at no cost with the rest of the world.
- Investment adjustment costs.
- No money or government. Real model.

The SOE-RBC model

- Preferences:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, h_t), \tag{1}$$

  where c_t is consumption, h_t is hours worked, $\beta \in (0, 1)$ is the subjective discount factor,
  $U_c > 0$, $U_h < 0$, $U_{cc} < 0$, $U_{hh} < 0$, $U_{ch} > 0$, and E_t denotes expectations conditional on information at time t.

- Period t budget constraint

$$y_t + d_t = c_t + i_t + d_{t-1}(1 + r_{t-1}) + \Phi(k_{t+1} - k_t). \tag{2}$$

  where y_t is output, i_t is investment, d_t is foreign debt at the end of period t, r_t is the interest rate, k_t is the stock of capital available for production at t (but chosen at $t - 1$).

- $\Phi(\cdot)$ represent investment costs, and satisfies: $\Phi(0) = 0$, $\Phi' > 0$, $\Phi'' > 0$ and $\Phi'(0) = 0$.

- Notice the timing convention for debt and capital is different.

The SOE-RBC model

- Technology:

$$y_t = A_t F(k_t, h_t), \tag{3}$$

where

$$\ln A_t = \rho \ln A_{t-1} + \eta \epsilon_t,$$

with ϵ_t i.i.d.$N(0,1)$, and $\rho \in [0,1)$.

  - Under this, $E_t\{\ln A_{t+j}\} = \rho^j \ln A_t$. Thus, A_t is expected to converge to 1.

- Investment:

$$k_{t+1} = i_t + (1-\delta)k_t, \tag{4}$$

with $\delta \in [0,1]$.

The SOE-RBC model

- Households choose processes $\{c_t, h_t, y_t, i_t, d_t, k_{t+1}\}_{t=0}^{\infty}$ to maximize (1) subject to (2)-(4) and the no-Ponzi game constraint

$$\lim_{j \to \infty} E_t \frac{d_{t+j}}{\prod_{s=0}^{j}(1+r_s)} \leq 0. \tag{5}$$

- To simplify, we can use (3) and (4) to eliminate y_t and i_t from (2) to get

$$A_t F(k_t, h_t) + d_t = c_t + k_{t+1} - (1-\delta)k_t + d_{t-1}(1+r_{t-1}) + \Phi(k_{t+1} - k_t). \tag{6}$$

The SOE-RBC model

- The Lagrangian is

$$E_0 \sum_{t=0}^{\infty} \beta^t \{ U(c_t, h_t) + \lambda_t [A_t F(k_t, h_t) + d_t \ldots$$
$$- c_t - k_{t+1} + (1-\delta) k_t - d_{t-1}(1 + r_{t-1}) - \Phi(k_{t+1} - k_t)] \} .$$

- Choices: $c_t, h_t, d_t, k_{t+1}, \lambda_t$.
- The first order conditions are (5) holding with equality, (6) and

$$\lambda_t = U_c(c_t, h_t), \tag{7}$$

$$-U_h(c_t, h_t) = \lambda_t A_t F_h(k_t, h_t) \tag{8}$$

$$\lambda_t = \beta(1 + r_t) E_t \lambda_{t+1} \tag{9}$$

$$\lambda_t [1 + \Phi'(k_{t+1} - k_t)] = \beta E_t \lambda_{t+1} [A_{t+1} F_k(k_{t+1}, h_{t+1}) + \ldots$$
$$1 - \delta + \Phi'(k_{t+2} - k_{t+1})] \tag{10}$$

Inducing stationarity

- In this model the marginal utility of consumption features a random walk; which also translates to all other variables.
- To see this, consider a simpler model like the following:

$$E_0 \sum_{t=0}^{\infty} \beta^t \, U(c_t),$$

subject to

$$y_t + d_t = d_{t-1}(1+r) + c_t.$$

where y_t is an exogenous endowment and r^* is the exogenous world interest rate (assumed constant for simplicity).

- The FOC is

$$U_{c,t} = \beta(1+r^*)E_t\{U_{c,t+1}\}.$$

- In SS, the equilibrium conditions are:

$$1 = \beta(1+r^*), \quad y = r^* d + c.$$

Inducing stationarity

- Thus, given that β and r^* are parameters, we need to impose a restriction on them: $\beta(1+r^*)=1$.
- The problem is that we are left with one equation $y=r^*d+c$ and two endogenous variables (c,d). So there are infinite possible SS.
- Another way to see that this is problematic: if $\beta(1+r^*)=1$, the equilibrium condition $U_{c,t}=\beta(1+r^*)E_t\{U_{c,t+1}\}$ implies that the marginal utility of consumption (and consumption in a log-linear approximation as we will see) will display a unit root.
- This is problematic because the solution techniques that we use requires stationarity of the variables. Thus, we will add some feature to induce stationarity of the solution.

Inducing stationarity

- The alternative that we will discuss here is called External Debt-Elastic Interest Rate (EDEIR). In particular, we assume that

$$r_t = r^* + p(\tilde{d}_t),$$

where r^* is the world interest rate (assumed constant for simplicity) and $p(\tilde{d}_t)$ is a country premium that is increasing in the evolution of the aggregate debt level of the economy $\tilde{d}_t$.

  - In equilibrium, because there is a representative agent, we have $d_t = \tilde{d}_t$. But the fact that the premium depends on the aggregate level implies that the household does not internalize that her decisions will change the premium (that is why we call this case "External").

- Why this premium helps to induce stationarity? If household would like to increase their level of debt after a given shock, the premium will rise making debt less appealing. Thus, the level of debt cannot be a random walk, because the premium will force debt to eventually return to its original level.

Equilibrium conditions

- In equilibrium, $d_t = \tilde{d}_t$. Eliminating λ_t we have the following equilibrium conditions:

$$A_t F(k_t, h_t) + d_t = c_t + k_{t+1} - (1-\delta)k_t + \ldots$$
$$d_{t-1}[1 + r^* + p(d_{t-1})] + \Phi(k_{t+1} - k_t) \quad (11)$$

$$-U_h(c_t, h_t) = U_c(c_t, h_t) A_t F_h(k_t, h_t) \quad (12)$$

$$U_c(c_t, h_t) = \beta[1 + r^* + p(d_t)] E_t[U_c(c_{t+1}, h_{t+1})] \quad (13)$$

$$U_c(c_t, h_t)[1 + \Phi'(k_{t+1} - k_t)] = \beta E_t U_c(c_{t+1}, h_{t+1}) \ldots$$
$$[A_{t+1} F_k(k_{t+1}, h_{t+1}) + 1 - \delta + \Phi'(k_{t+2} - k_{t+1})] \quad (14)$$

$$\lim_{j\to\infty} E_t \frac{d_{t+j}}{\prod_{s=0}^{j}(1 + r_s)} = 0 \quad (15)$$

$$\ln A_t = \rho \ln A_{t-1} + \eta \epsilon_t \quad (16)$$

Equilibrium conditions

- A stationary equilibrium is a set of processes $\{d_t, c_t, h_t, k_{t+1}, A_t\}_{t=0}^{\infty}$ satisfying (11)-(16), given A_0, d_{-1}, k_0 and a process for $\{\epsilon_t\}_{t=0}^{\infty}$.
- Additionally we can define

$$y_t = A_t F(k_t, h_t), \tag{17}$$

$$i_t = k_{t+1} - (1-\delta)k_t, \tag{18}$$

$$tb_t = y_t - c_t - i_t - \Phi(k_{t+1} - k_t) \tag{19}$$

$$ca_t = d_{t-1} - d_t = tb_t - r_{t-1}d_{t-1} \tag{20}$$

Decentralization

- In this description of the equilibrium, because there is a single agent that controls everything, there are many relevant prices that are not defined (e.g. real wages, rental rates of capital, price of capital, etc.)
- We could instead decentralize the equilibrium in several alternative ways with competitive markets.
- But because the assumptions for the welfare theorems hold in this model, the equilibrium for the variables that appear in both the centralized and in the decentralized equilibria (c_t, h_t, k_t, etc.) will be the same.

Functional Forms

- Instantaneous utility

$$U(c,h) = \frac{\left[c - \frac{h^\omega}{\omega}\right]^{1-\sigma} - 1}{1-\sigma}$$

These are known as GHH preferences. In particular, they imply that

$$\frac{-U_h(c,h)}{U_c(c,h)} = h^{\omega-1}.$$

which is independent from consumption. Thus labor supply features no wealth effect.

- Production Function

$$F(k,h) = k^\alpha h^{1-\alpha}$$

with $\alpha \in (0,1)$

Functional Forms

- Capital adjustment cost

$$\Phi(x) = \frac{\phi}{2}(x)^2,$$

with $\phi > 0$.

- Premium

$$p(d) = \psi_1 \left(e^{d-\bar{d}} - 1 \right)$$

where $\psi_1 > 0$ and $\bar{d}$ are parameters.

On the Solution of the Model

- The equilibrium conditions form a dynamic and stochastic system of non-linear difference equations; whose algebraic solution will generally not be available (like the standard RBC model discussed before).
- Instead, we will find an approximate solution, which we will be able to compute numerically (i.e. for a given value of parameters).
- To do this, we need:
    - To find the non-stochastic steady-state
    - To assign values for the parameters
    - To solve the dynamic and stochastic system of linear difference equations

Steady State

The non-stochastic steady-state refers to the equilibrium where there are no shocks ($\epsilon_t = 0$ for all t). The steady state conditions are

$$Ak^{\alpha}h^{1-\alpha}+d = c+k-(1-\delta)k+d\left[1+r^{*}+\psi_1\left(e^{d-\bar{d}}-1\right)\right]+\frac{\phi}{2}(k-k)^2.$$

$$h^{\omega-1} = A(1-\alpha)k^{\alpha}h^{-\alpha}$$

$$\left[c-\frac{h^{\omega}}{\omega}\right]^{-\sigma} = \beta\left[1+r^{*}+\psi_1\left(e^{d-\bar{d}}-1\right)\right]\left[c-\frac{h^{\omega}}{\omega}\right]^{-\sigma}$$

$$\left[c-\frac{h^{\omega}}{\omega}\right]^{-\sigma}\left[1+\phi(k-k)\right] = \beta\left[c-\frac{h^{\omega}}{\omega}\right]^{-\sigma}\left[A\alpha k^{\alpha-1}h^{1-\alpha}+1-\delta+\phi(k-k)\right]$$

$$\ln A = \rho \ln A$$

Steady State

- If we assume that $\beta(1+r^*) = 1$, and simplifying,

$$Ak^{\alpha}h^{1-\alpha} = c + \delta k + dr^* \tag{21}$$

$$h^{\omega-1} = A(1-\alpha)k^{\alpha}h^{-\alpha} \tag{22}$$

$$d = \bar{d} \tag{23}$$

$$1 = \beta\left[A\alpha k^{\alpha-1}h^{1-\alpha} + 1 - \delta\right] \tag{24}$$

$$A = 1 \tag{25}$$

Steady State

- Using (24)

$$1 = \beta \left[\alpha \left(\frac{k}{h} \right)^{\alpha - 1} + 1 - \delta \right], \Rightarrow \kappa \equiv \frac{k}{h} = \left(\frac{\beta^{-1} - 1 + \delta}{\alpha} \right)^{\frac{1}{\alpha - 1}}$$

- From (22)

$$h = [(1 - \alpha)\kappa^{\alpha}]^{\frac{1}{\omega - 1}}$$

- Then

$$k = \kappa h, \;\; c = k^{\alpha} h^{1-\alpha} - r^* \bar{d} - \delta k$$

Calibration Strategy

- Parameters of the model: $\sigma, \beta, \omega, \alpha, \delta, \phi, \psi_1, r^*, \rho, \eta, \bar{d}$.
- Divide the parameters in 3 groups
    - Parameters calibrated sourced unrelated to the data the model aims to explain.
    - Parameters set to match first moments of the data the model aims to explain.
    - Parameters set to match second moments of the data the model aims to explain.
- Data aimed to explain: Canada (Mendoza, 1991). The time unit is meant to be a year.

Calibration Strategy

- Parameters calibrated sourced unrelated to the data the model aims to explain.
  - In this case: σ, δ. Chosen values: $\sigma = 2$, $\delta = 0.1$.
- Parameters set to match first moments of the data the model aims to explain.
  - α is set to match the average labor share in total GDP ($1-\alpha$ in the model).
  - r^* is set to match the average interest rate. Notice that, given $\beta(1+r^*) = 1$, we obtain β as well.
  - $\bar{d}$: in the model, we have $tb = r^* \bar{d}$. Then, we set $\bar{d}$ to match the average tb/y ratio.
  - We will later argue that the steady state is also the unconditional mean of the model given the solution method we will use.

Calibration Strategy

- Parameters set to match second moments of the data the model aims to explain.
  - $\omega, \phi, \psi_1, \rho, \eta$ are set to match the second moments:
    - Standard deviations: hours worked (2.02 %), investment (9.82 %), trade-balance-to-output ratio (1.87 %), output (2.81 %).
    - Autocorrelation of output (0.62).

Calibration Result

- Given our strategy, we are left with the following parameters:

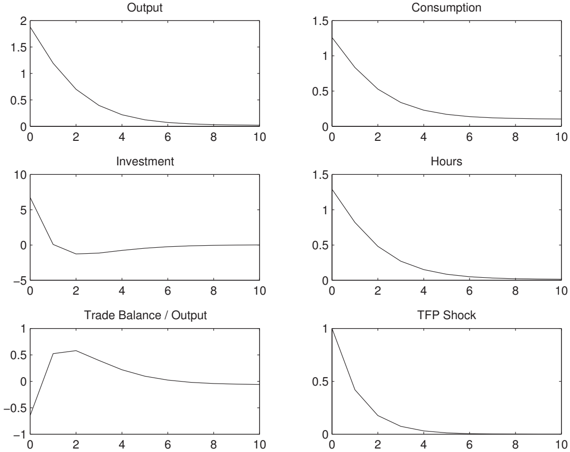
σ	β	ω	α	δ	ϕ
2	0.96	1.455	0.32	0.1	0.028
ψ_1	r^*	ρ	η	$\bar{d}$	
0.000742	0.04	0.42	0.0129	0.7442	

Computational implementation

- Toolkit developed by Uribe and Schmitt-Grohe: `http://www.columbia.edu/ mu2166/1st_order/1st_order.htm`
- Dynare: `http://www.dynare.org` (can also be implemented in Octave, an open-source and free version of Matlab).

Performance of the Model

Figure 4.1: Responses to a One-Percent Productivity Shock



Note. To produce this figure, run the Matlab code edeir_run.m.

Performance of the Model

Table 4.2: Empirical and Theoretical Second Moments

Variable	Canadian Data 1946 to 1985			Canadian Data 1960 to 2011			Model		
	σ_{x_t}	$\rho_{x_t,x_{t-1}}$	ρ_{x_t,GDP_t}	σ_{x_t}	$\rho_{x_t,x_{t-1}}$	ρ_{x_t,GDP_t}	σ_{x_t}	$\rho_{x_t,x_{t-1}}$	ρ_{x_t,GDP_t}
y	2.81	0.62	1	3.71	0.86	1	3.08	0.62	1
c	2.46	0.70	0.59	2.19	0.70	0.62	2.71	0.78	0.84
i	9.82	0.31	0.64	10.31	0.69	0.80	9.04	0.07	0.67
h	2.02	0.54	0.80	3.68	0.75	0.78	2.12	0.62	1
$\frac{tb}{y}$	1.87	0.66	-0.13	1.72	0.76	0.12	1.78	0.51	-0.04
$\frac{ca}{y}$							1.45	0.32	0.05

Note. Empirical moments for the peirod 1946 to 1985 are taken from Mendoza (1991) and for the period 1960 to 2011 are based on own calculations using data from WDI (GDP, consumption, investment, imports, and exports) and Statistics Canada (hours worked). All empirical second moments based on annual, per capita, and quadratically detrended data. Standard deviations are measured in percentage points. Theoretical moments are produced by running the Matlab code edeir_run.m.

Performance of the Model

Figure 4.2: Response of the Trade-Balance-To-Output Ratio to a Positive Technology Shock